Flood Probability Prediction Model Using Machine Learning and Deep Learning

Pragga Falguni Roy Chowdhury[1] Sudipta Goswami[2] Shahriar[3] Sabrina Sultana[4]
Dept. of Computer Science and Engineering, BRAC University, Dhaka, Bangladesh
[1]praggafalguni.roy.chowdhury@g.bracu.ac.bd [2]sudipta.goswami@g.bracu.ac.bd
[3]shahriar@g.bracu.ac.bd [4]sabrina.sultana@g.bracu.ac.bd

***Abstract*—Floods are among the most destructive natural disasters, causing loss of life, infrastructure damage, displacement, and economic disruption. Accurate flood probability prediction can support early warning, preparedness planning, and resource allocation. This report presents a regression-based comparative study of machine learning and neural-network models for predicting continuous flood probability. The implementation uses a dataset of 50,000 records with 20 explanatory features representing meteorological, hydrological, geographical, infrastructural, environmental, and policy-related factors. Seven models are implemented and evaluated: K-Nearest Neighbors (KNN), custom gradient descent linear regression, Random Forest, XGBoost, Gradient Boosting, AdaBoost, and Multi-layer Perceptron (MLP). Because the target variable is continuous, the evaluation uses Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and coefficient of determination (R^2). The custom gradient descent model achieved the best test performance with MSE of 4.71×10^{-10}, MAE of 0.000022, and $R^2 \approx 1.000000$, followed by the MLP model with $R^2 = 0.989839$ and XGBoost with $R^2 = 0.925209$. The results indicate that the dataset has a strong linear or additive feature-target structure. The report further includes correlation analysis, feature importance, residual plots, actual-versus-predicted visualizations, cross-validation, hyperparameter tuning, paired statistical testing, outlier analysis, tolerance accuracy, and KMeans clustering with PCA visualization.**

***Index Terms*—Flood prediction, machine learning, regression, gradient descent, MLP, XGBoost, Random Forest, KNN, disaster management.**

I. Introduction

Floods are among the most damaging natural hazards, producing severe social, economic, environmental, and infrastructural impacts. As climate variability intensifies, land-use patterns shift, urban expansion accelerates, and drainage systems deteriorate, reliable flood prediction systems have become increasingly important for disaster preparedness, emergency response, and long-term planning.

Traditional flood forecasting often relies on hydrological and hydraulic simulations. Such physics-based models are valuable because they represent rainfall-runoff and flow processes explicitly; however, they often require detailed basin-level calibration data, expert parameterization, and considerable computational resources. These requirements can limit operational scalability, particularly in data-scarce regions.

Machine learning (ML) and deep learning (DL) offer complementary data-driven approaches. ML models can learn relationships between flood probability and explanatory variables such as monsoon intensity, topographic drainage, river management, urbanization, climate change, infrastructure condition, wetland loss, and disaster preparedness. Unlike fully physical models, ML systems can be retrained when new observations become available and can support rapid inference once trained.

This study formulates flood forecasting as a supervised regression problem. The continuous target variable, *FloodProbability*, is predicted from 20 explanatory variables. Seven regression models are implemented and compared: KNN Regressor, custom gradient descent linear regression, Random Forest Regressor, XGBoost Regressor, Gradient Boosting Regressor, AdaBoost Regressor, and MLP Regressor. The work also includes visual and statistical diagnostics to verify whether the numerical metrics are consistent with model behavior.

II. Literature Review

A. Hydrological, Statistical, and Data-Driven Flood Prediction

Flood prediction research has evolved from statistical and physics-based methods toward data-driven, hybrid, and interpretable learning systems. Systematic reviews show that traditional hydrological models such as 1D, 2D, and 3D flood models provide physical interpretability but often involve high calibration effort and computational cost [1], [7]. Earlier time-series approaches, including ARMA and ARIMA, provided structured rainfall forecasting baselines but struggled with nonlinear flood dynamics [9], [10].

The increasing availability of environmental datasets has shifted attention toward ML-based methods. Reviews and comparative studies report that algorithms such as Artificial Neural Networks (ANNs), Support Vector Machines (SVMs), KNN, Random Forest, Gradient Boosting, XGBoost, and MLPs can capture nonlinear relationships between environmental predictors and flood outcomes [1], [7]. However, model quality depends strongly on training data quality, feature engineering, validation strategy, and whether the prediction task is framed as classification, susceptibility mapping, water-level forecasting, discharge prediction, or continuous probability regression.

B. Ensemble Learning and Hybrid Modeling

Ensemble models are widely used in flood susceptibility and forecasting because they can model nonlinear interactions and handle heterogeneous features. Random Forest reduces variance through bagging, while XGBoost and Gradient Boosting correct residual errors sequentially through boosting. A Yellow River basin study combined k-means clustering, random forest pattern recognition, and the Xin'anjiang hydrological model to improve real-time parameter migration, showing the benefit of linking ML with hydrological domain structure [2]. Similarly, ensemble methods such as Random Forest, LightGBM, and CatBoost have produced highly accurate flood susceptibility maps when combined with physical hydrological modeling [5]. XGBoost has also been used in data-driven flood alert systems for short-term stage forecasting at multiple lead times [14].

Hybrid ML-hydraulic systems are especially important because pure ML predictions may not directly provide inundation extent, depth, or hazard maps. Studies coupling LSTM predictions with HEC-RAS 2D hydraulic modeling demonstrate that learned streamflow forecasts can be converted into spatial flood risk maps for decision-making [17]. This reinforces the role of ML as a component within a broader operational flood-risk workflow rather than as a complete replacement for hydrological modeling.

C. Deep Learning and Interpretability

Deep learning has become prominent in flood forecasting, especially for time-series tasks. LSTM, Bidirectional LSTM, Deep Recurrent Neural Networks, and attention-based LSTM variants have been used for daily or hourly rainfall-runoff and water-level prediction [6], [13]. Explainable LSTM modules and physics-informed attention networks address the black-box nature of deep learning by identifying influential variables and time steps [8], [16]. These studies demonstrate that neural networks can be accurate and interpretable when paired with attention, feature attribution, or physical constraints.

The present implementation uses an MLP rather than a convolutional or recurrent architecture because the dataset is structured tabular data rather than image, raster, or temporal sequence data. MLPs are appropriate for tabular regression when features are scaled and the dataset contains sufficient examples. However, for future real-time flood forecasting with rainfall sequences, river gauge time series, or satellite image stacks, LSTM, temporal transformer, CNN-LSTM, or physics-informed architectures would be more suitable.

D. Remote Sensing, Bangladesh Context, and Climate Adaptation

Remote sensing and cloud-based geospatial platforms have expanded flood prediction in data-scarce regions. Studies using Sentinel imagery, Google Earth Engine, SAR products, U-Net segmentation, and SHAP analysis show that geospatial flood inventories and conditioning factors can support scalable susceptibility mapping [3], [19]. Bangladesh-focused studies emphasize that river water-level forecasting requires careful treatment of spatial variation, upstream controls, and data limitations. Random Forest, XGBoost, LightGBM, MLP, and attention-based LSTM models have been evaluated for the Old Brahmaputra, Dhaka, and Sylhet contexts, showing strong potential for operational forecasting when reliable water-level histories are available [11]–[13].

Recent work also extends flood prediction toward climate resilience and adaptive planning. RAPFLO uses hierarchical deep neural networks for rapid urban flood risk prediction [4]. Cascade machine learning frameworks have been used for probabilistic flood risk assessment under climate projections [18], atmospheric-circulation classification has been applied to compound flood risk assessment [20], and reinforcement learning has been used for adaptive coastal flood-risk management policies [21]. These studies indicate that future flood systems should combine prediction, uncertainty estimation, spatial mapping, and adaptation planning.

E. Positioning of the Present Study

The present study differs from many time-series and satellite-based flood studies because it uses a structured tabular dataset with 20 static explanatory factors and a continuous target variable. Therefore, the correct modeling formulation is regression, not classification. This report evaluates classical ML models, a custom gradient descent implementation, and an MLP regressor under a consistent preprocessing and evaluation pipeline. The main contribution is a clear code-aligned comparison of regression models with diagnostics that explain why the best model performs so strongly.

III. Methodology

A. Dataset Description

The dataset contains 50,000 records and 21 columns: 20 numeric input features and one continuous target variable, *FloodProbability*. The explanatory features are Monsoon Intensity, Topography Drainage, River Management, Deforestation, Urbanization, Climate Change, Dams Quality, Siltation, Agricultural Practices, Encroachments, Ineffective Disaster Preparedness, Drainage Systems, Coastal Vulnerability, Landslides, Watersheds, Deteriorating Infrastructure, Population Score, Wetland Loss, Inadequate Planning, and Political Factors.

All features are numeric. The implementation checks dataset shape, data types, descriptive statistics, unique values, and missing values. No missing values are present, and duplicate removal does not change the dataset size. Although the preprocessing pipeline includes `fillna(df.mean())` as a defensive step, it has no practical effect on this dataset.

B. Exploratory Data Analysis

EDA is performed to inspect feature distributions, correlation patterns, and the relationship between predictors and the target. Fig. 1 shows the Pearson correlation heatmap across all variables. The heatmap indicates that correlations among predictors are generally weak to moderate, and no single variable dominates the target relationship. This supports the interpretation that *FloodProbability* is produced by the combined contribution of multiple factors.

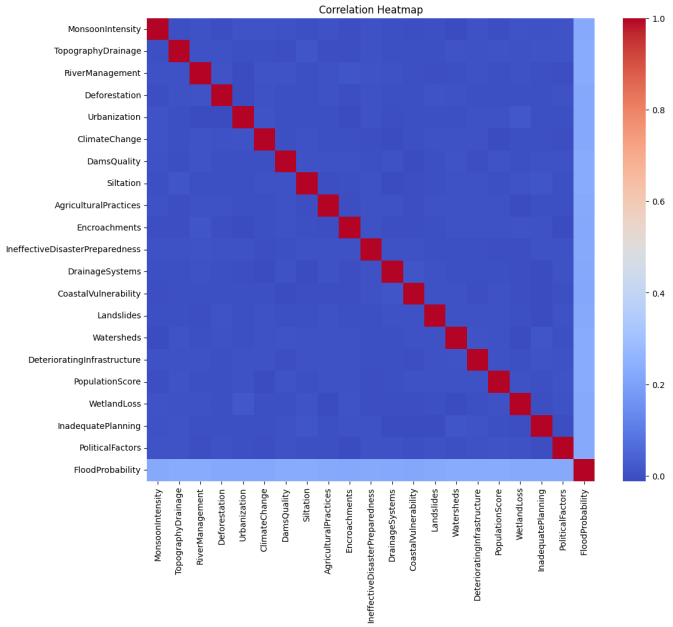


Fig. 1. Correlation heatmap generated from the implementation notebook. The heatmap supports the observation that the target is influenced by multiple predictors rather than one dominant feature.

C. Preprocessing Pipeline

The feature matrix and target vector are separated as

$$X = \text{all columns except FloodProbability}, y = \text{FloodProbability} \tag{1}$$

The dataset is split into training and test sets using an 80/20 split with `random_state=42`, producing 40,000 training samples and 10,000 test samples.

Feature scaling is applied selectively. KNN, gradient descent, and MLP are trained on standardized features because they are sensitive to feature magnitudes. Standardization uses

$$z = \frac{x - \mu}{\sigma}, \tag{2}$$

where μ is the training-set mean and σ is the training-set standard deviation. Tree-based models are trained on the original unscaled features because decision-tree split rules are not dependent on monotonic feature scaling.

D. Implemented Models

1) K-Nearest Neighbors Regressor: KNN predicts the output of a test sample by averaging the target values of the $k = 5$ nearest training samples in standardized feature space. It is non-parametric and can capture local patterns, but performance can degrade with high-dimensional data.

2) Custom Gradient Descent Linear Regression: A linear regression model is implemented manually using batch gradient descent. A bias term is prepended to the scaled feature matrix, producing a bias-augmented matrix X_b. The parameter vector θ is initialized to zero and updated for 1000 epochs with learning rate $\alpha = 0.01$:

$$\theta \leftarrow \theta - \frac{\alpha}{m} X_b^T (X_b\theta - y), \tag{3}$$

where m is the number of training samples. Test predictions are computed as

$$\hat{y} = X_{b,test}\theta. \tag{4}$$

3) Random Forest Regressor: Random Forest constructs an ensemble of decision trees using bootstrap aggregation. Each tree is trained on a bootstrapped sample and considers a random subset of features at each split. Final predictions are obtained by averaging tree outputs.

4) XGBoost Regressor: XGBoost is a regularized gradient boosting algorithm using the squared-error objective. It builds trees sequentially, with each tree correcting the residual errors of the current ensemble. Regularization helps control model complexity.

5) Gradient Boosting Regressor: Gradient Boosting also builds trees sequentially on residuals, but without all the regularization and optimization features available in XGBoost. It provides a strong boosting baseline.

6) AdaBoost Regressor: AdaBoost iteratively trains weak learners while increasing emphasis on samples with higher prediction error. It is useful as a simple boosting comparison model.

7) Multi-layer Perceptron Regressor: The neural-network model is an MLP Regressor with two hidden layers of 100 and 50 neurons. It is trained on standardized inputs for a maximum of 500 iterations. The implementation uses the default ReLU activation and Adam optimizer from scikit-learn.

E. Evaluation Metrics

Because *FloodProbability* is continuous, regression metrics are used. For n test samples, actual values y_i, predictions $\hat{y}_i$, and target mean $\bar{y}$, the metrics are:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2, \tag{5}$$

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}, \tag{6}$$

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|, \tag{7}$$

$$\text{MAPE} = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right|, \tag{8}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}. \tag{9}$$

Lower MSE, RMSE, MAE, and MAPE indicate lower error. An R^2 value closer to 1 indicates stronger predictive fit.

F. Diagnostic Analysis

The implementation includes additional diagnostics beyond the primary metrics. Residual plots are generated for all seven models by plotting $e_i = y_i - \hat{y}_i$ against predicted values. Actual-versus-predicted plots compare predicted values against the perfect prediction line. Five-fold cross-validation evaluates Random Forest stability, while GridSearchCV performs three-fold tuning over `n_estimators`, `max_depth`,

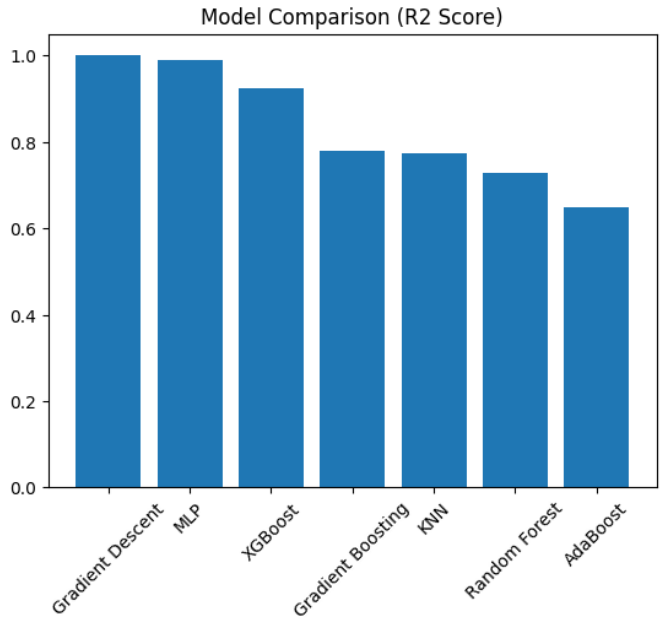


Fig. 2. Model comparison using R^2 score. Gradient Descent, MLP, and XGBoost are the top three models.

and `min_samples_split`. A paired t-test compares the prediction arrays of the two top-ranked models. Random Forest absolute errors are also used for outlier detection, with samples above the 95th percentile flagged as outliers. Finally, KMeans clustering with $k = 3$ and PCA visualization explores natural structure in the feature space.

IV. Results and Discussion

A. Model Performance Comparison

Table I presents the test-set performance of all seven models, sorted by R^2. Fig. 2 visualizes the same ranking using R^2.

The custom gradient descent model achieves the strongest result with $R^2 \approx 1.000000$ and extremely low prediction error. This indicates that the target variable is almost perfectly represented by a linear or additive function of the 20 input features in this dataset. The MLP ranks second, showing that a feedforward neural network can approximate the target function closely. XGBoost is the best tree-based ensemble model, with $R^2 = 0.925209$. Gradient Boosting and KNN show moderate performance, while Random Forest and AdaBoost underperform relative to the top three models.

B. Interpretation of the Best Model

The custom gradient descent model implements batch linear regression through iterative parameter updates. Its near-perfect fit indicates that *FloodProbability* is approximately a linear combination of the 20 input features in this dataset.

The near-perfect performance of the gradient descent model also suggests that the dataset may have a synthetic or highly controlled structure. Real-world flood systems usually involve nonlinear, time-dependent, and spatially complex relationships among rainfall, drainage, river flow, land use, and infrastructure conditions; therefore, a nearly perfect linear fit strongly implies that *FloodProbability* may have been generated as a weighted additive combination of the input features rather than observed from real flood events [1], [7].

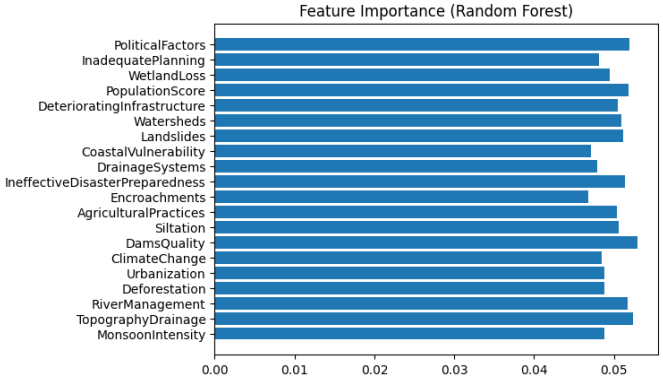


Fig. 3. Random Forest feature importance. The relatively balanced bars indicate that prediction depends on multiple features rather than one dominant variable.

Before applying this model operationally, it should be validated on real-world or external flood datasets to confirm that the linear relationship generalizes beyond the training distribution.

C. Feature Importance

Fig. 3 presents Random Forest feature importance. The feature importance values are relatively close to one another, meaning no single predictor overwhelmingly dominates the prediction task. This is consistent with the correlation heatmap and with the multi-causal nature of flood probability. Governance, infrastructure, drainage, land use, and environmental features collectively shape the model output.

D. Residual Analysis

Fig. 4 shows residual plots for all seven models. Gradient Descent residuals are extremely small, on the order of 10^{-5}, although the zoomed scale reveals a slight systematic trend. This trend is numerically negligible compared with the overall target range and likely reflects finite gradient descent convergence after 1000 epochs. MLP residuals are also tightly centered near zero. XGBoost has a wider but still controlled residual spread. KNN, Gradient Boosting, Random Forest, and AdaBoost show progressively larger dispersion, consistent with their lower R^2 values.

E. Actual Versus Predicted Analysis

Fig. 5 shows actual-versus-predicted plots for all models. Points close to the diagonal line indicate strong agreement between predicted and actual flood probability. Gradient Descent almost perfectly overlaps with the diagonal. MLP and XGBoost also show strong alignment, with increasing scatter as performance decreases. Random Forest and AdaBoost show the largest deviations from the perfect prediction line, matching their weaker quantitative results.

TABLE I
REGRESSION PERFORMANCE OF THE IMPLEMENTED MODELS

Model	**MSE**	**RMSE**	**MAE**	**MAPE**	R^2
Gradient Descent	4.71×10^{-10}	0.000022	0.000022	0.000043	1.000000
MLP	2.53×10^{-5}	0.005031	0.003912	0.007909	0.989839
XGBoost	1.86×10^{-4}	0.013648	0.010814	0.021691	0.925209
Gradient Boosting	5.51×10^{-4}	0.023463	0.018481	0.037506	0.778957
KNN	5.61×10^{-4}	0.023690	0.018590	0.036885	0.774659
Random Forest	6.75×10^{-4}	0.025981	0.020462	0.041209	0.728976
AdaBoost	8.72×10^{-4}	0.029534	0.023420	0.047939	0.649784

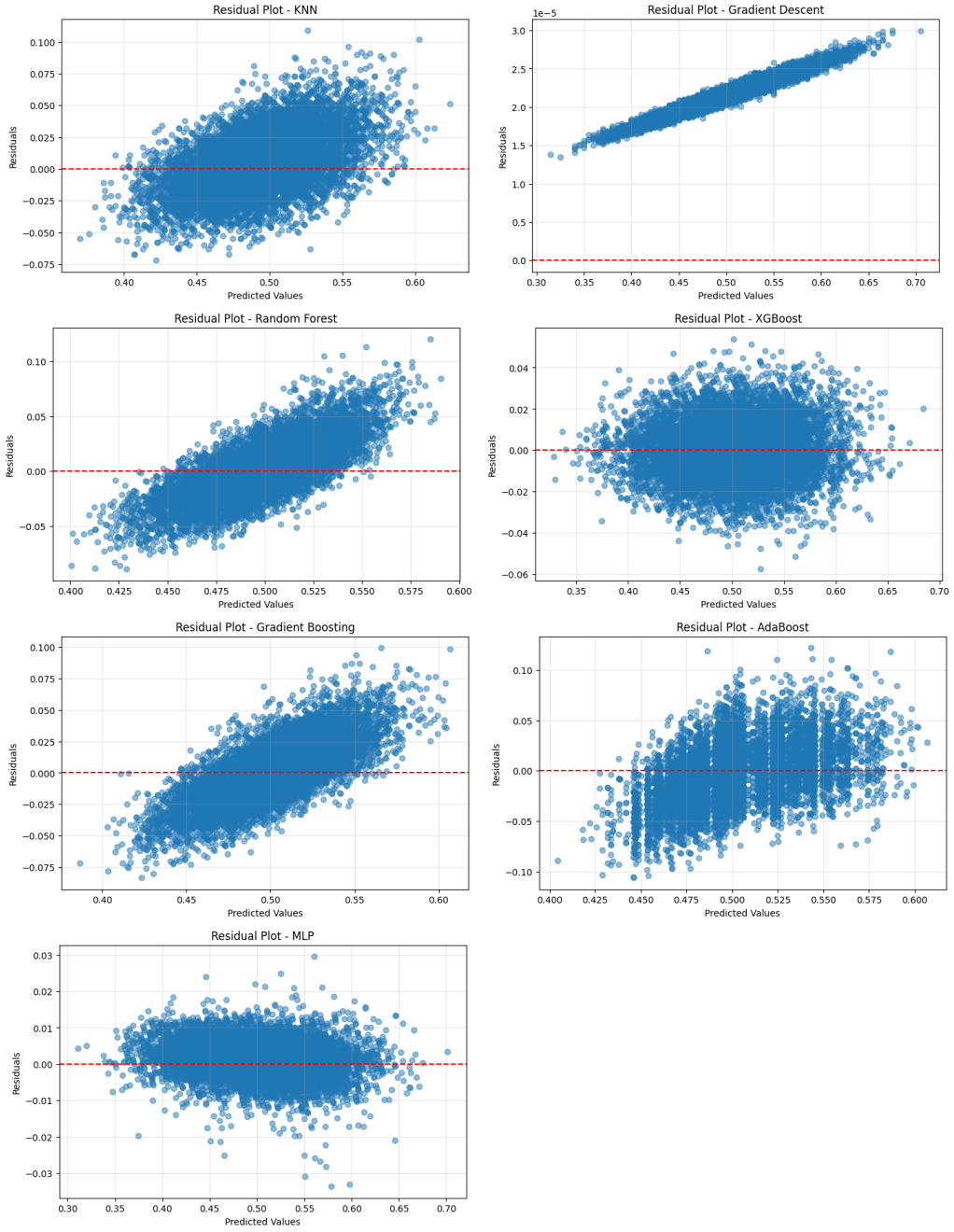


Fig. 4. Residual plots for all seven implemented models: KNN, Gradient Descent, Random Forest, XGBoost, Gradient Boosting, AdaBoost, and MLP. The titles inside each plot identify the corresponding model.

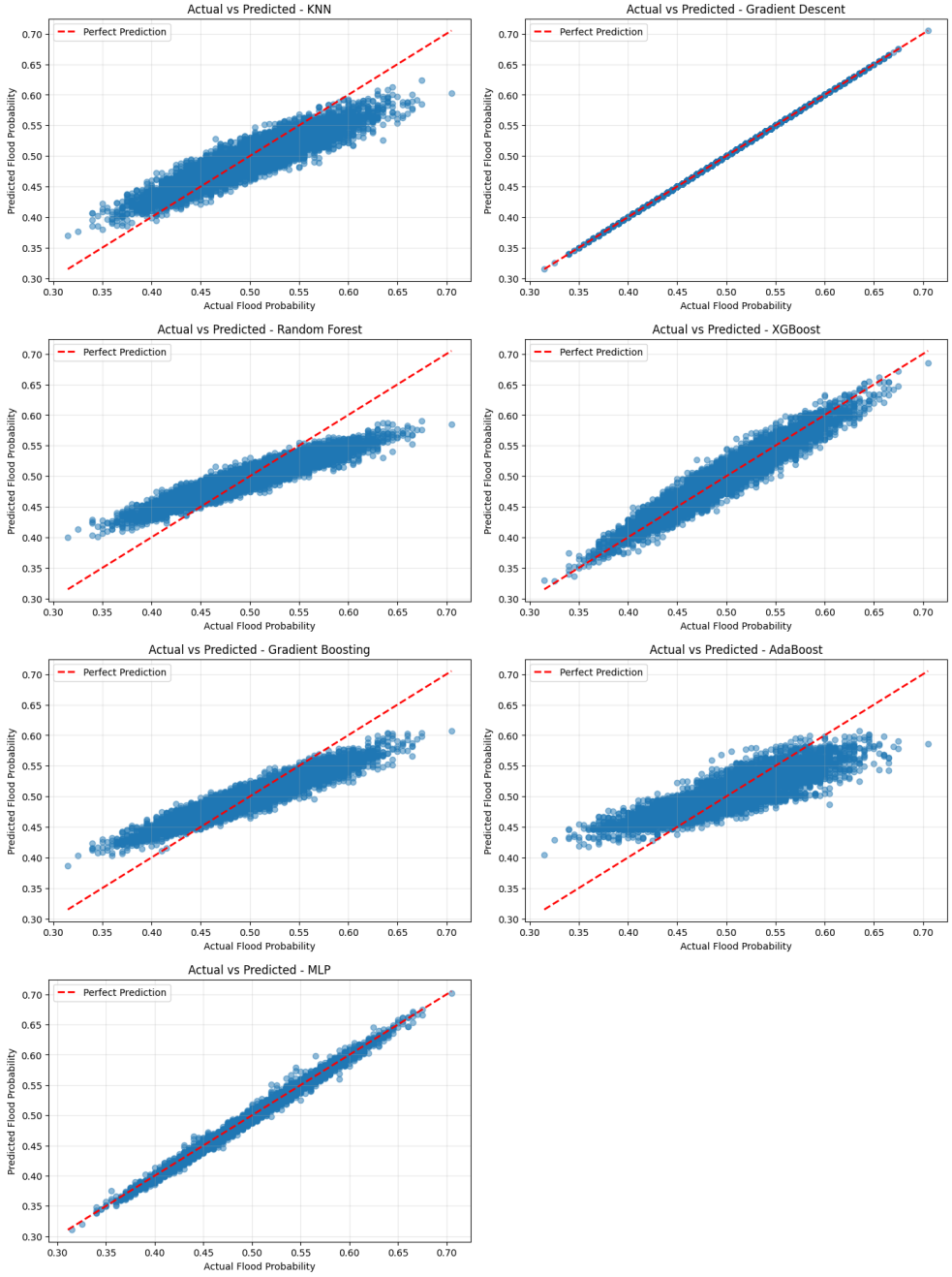


Fig. 5. Actual-versus-predicted plots for all seven implemented models. Diagonal alignment indicates stronger predictive agreement.

F. Cross-Validation, Hyperparameter Tuning, and Statistical Testing

Table II summarizes the additional validation and diagnostic outputs. Random Forest achieved a 5-fold cross-validation R^2 of 0.732657, close to its test R^2 of 0.728976, suggesting stable generalization. GridSearchCV selected 200 trees, no maximum depth, and minimum split size of 2 as the best Random Forest configuration.

The paired t-test was computed between the raw prediction arrays of Gradient Descent and MLP across 10,000 test samples. The resulting p-value is approximately zero, indicating a statistically significant difference between the two prediction arrays. However, this test should not be overinterpreted as a complete performance significance test because it compares prediction values directly. A paired test on absolute error arrays, squared error arrays, or fold-wise validation scores would be a more direct future comparison.

G. Tolerance Accuracy and Outlier Analysis

Tolerance accuracy was computed at threshold $\delta = 0.1$ as the proportion of samples with $|y_i - \hat{y}_i| \leq 0.1$. All models

TABLE II
ADDITIONAL VALIDATION AND DIAGNOSTIC OUTPUTS

Diagnostic Item	Result
Random Forest 5-fold CV R^2	0.732657
Best RF tuning configuration	200 trees, no max depth, min split 2
Best model	Gradient Descent
Second-best model	MLP
Paired t-test statistic	30.2751
Paired t-test p-value	≈ 0.000000
Random Forest outliers	500 samples above 95th percentile error

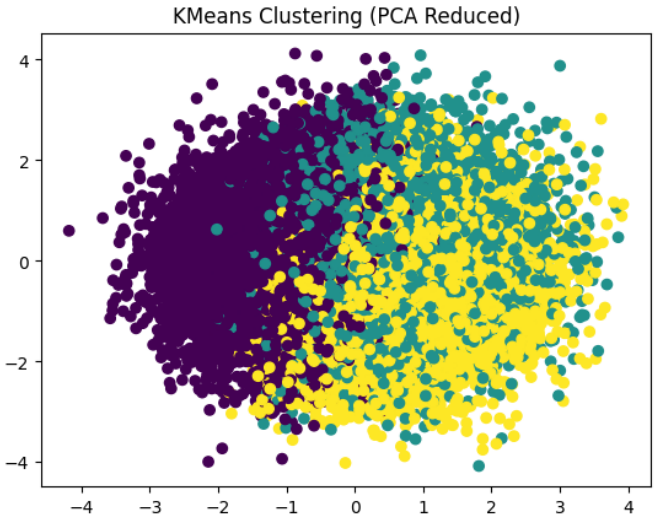


Fig. 6. KMeans clustering visualized after PCA reduction to two dimensions. The three clusters overlap partially, indicating moderate natural grouping in the feature space.

scored very high: KNN 0.9998, Gradient Descent 1.0000, Random Forest 0.9996, XGBoost 1.0000, Gradient Boosting 1.0000, AdaBoost 0.9987, and MLP 1.0000. Because the target range is narrow, a 0.1 tolerance is too permissive to meaningfully separate the models. The standard regression metrics in Table I are more informative.

Outlier analysis was performed using Random Forest absolute errors. Samples exceeding the 95th percentile of error were flagged, producing 500 outlier samples. These cases represent observations where Random Forest had comparatively higher difficulty and could be useful for future error analysis.

H. Clustering-Based Exploratory Analysis

KMeans clustering with $k = 3$ was applied to the full standardized feature matrix. PCA reduced the 20-dimensional feature space to two components for visualization, shown in Fig. 6. The plot shows three partially overlapping clusters. This indicates some natural grouping in feature profiles, but the boundaries are not sharply separated. This clustering analysis is exploratory and independent of supervised model training.

V. KEY IMPLEMENTATION CORRECTIONS

The updated code and report require several important corrections compared with the earlier classification-oriented version. First, the project must be described as a regression task because *FloodProbability* is continuous. Therefore, Accuracy, Precision, Recall, F1-score, ROC curve, and AUC are not the primary evaluation metrics. Second, the neural-network model is an MLP Regressor, not a true CNN, because the implementation uses fully connected layers and does not perform convolution. Third, the train-test split is 80/20 rather than 25/75. Fourth, the custom gradient descent regression model is now part of the model comparison and is the best-performing model. Fifth, residual plots and actual-versus-predicted plots are generated for all seven models and are included in this report.

VI. CONCLUSION

This study presents a regression-based comparison of machine learning and neural-network models for flood probability prediction using a structured tabular dataset of 50,000 records and 20 explanatory features. The custom gradient descent linear regression model achieved the best result with $R^2 \approx 1.000000$, MSE of 4.71×10^{-10}, and MAE of 0.000022. The MLP model also performed strongly with $R^2 = 0.989839$, while XGBoost was the best tree-based ensemble with $R^2 = 0.925209$.

The results suggest that the dataset has a strong linear or additive target structure. This finding explains why a simple gradient descent linear model outperformed more flexible nonlinear learners. Residual plots, actual-versus-predicted visualizations, feature importance, cross-validation, tuning outputs, statistical testing, outlier analysis, and clustering all support the main quantitative findings.

However, the near-perfect result should be treated with caution. Real flood behavior is affected by temporal rainfall patterns, upstream river flow, soil saturation, sedimentation, drainage performance, land-use change, and institutional response. Future work should validate the model on external real-world data, incorporate river gauge and rainfall time series, integrate satellite-based flood maps, compare temporal deep learning architectures, and add explainability methods such as SHAP to improve operational trust.

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